Using SE formulas

MATH 213

Group 1 - Question 1

In a random sample of 765 adults in the United States, 322 say they could not cover a $400 unexpected expense without borrowing money or going into debt. (We have seen this example before.)

Estimate the standard error for this kind of random sample:

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.

Interpret your 50% confidence interval in the usual way, using the context:

Question 2

Another example we have been using is about estimating the mean income in a population using the mean income in a random sample of 500 incomes from that population. The sample mean income is $43800, and the sample standard deviation is $5830.

Estimate the standard error for this kind of random sample:

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.

Interpret your 50% confidence interval in the usual way, using the context:

Group 2 - Question 1

In a random sample of 765 adults in the United States, 322 say they could not cover a $400 unexpected expense without borrowing money or going into debt. (We have seen this example before.)

Estimate the standard error for this kind of random sample:

0.0178

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.

$0.42 - (\frac{2}{3} * 0.0178) = 0.408$

$0.42 + (\frac{2}{3} * 0.0178) = 0.432$

50% CI: [0.408, 0.432]

Interpret your 50% confidence interval in the usual way, using the context:

We are 50% confident that between 40.8% and 43.2% of adults in the U.S. could not cover a $400 unexpected expense without borrowing money or going into debt.

Question 2

Another example we have been using is about estimating the mean income in a population using the mean income in a random sample of 500 incomes from that population. The sample mean income is $43800, and the sample standard deviation is $5830.

Estimate the standard error for this kind of random sample:

260.726

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.

$43800 - (\frac{2}{3} * 260.726) = 43626.18$

$43800 + (\frac{2}{3} * 260.726) = 43973.82$

50% CI: [43626.18, 43973.82]

Interpret your 50% confidence interval in the usual way, using the context:

We are 50% confident that the mean income in this population is between $43626.18 and $43973.82.

Group 3 - Question 1

In a random sample of 765 adults in the United States, 322 say they could not cover a $400 unexpected expense without borrowing money or going into debt. (We have seen this example before.)

Estimate the standard error for this kind of random sample:

sqrt((0.421(1-0.421)) / 765) = 0.0178

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.

$.421 + \frac{2}{3} \times 0.0178 = .432867$

$.421 - \frac{2}{3} \times 0.0178 = .409133$

Interpret your 50% confidence interval in the usual way, using the context:

We are 50% confidence that between 40.9% and 43.2% of all americans could not cover a $400 unexpected expense without borrowing money or going into debt.

Question 2

Another example we have been using is about estimating the mean income in a population using the mean income in a random sample of 500 incomes from that population. The sample mean income is $43800, and the sample standard deviation is $5830.

Estimate the standard error for this kind of random sample:
5830 / sqrt(500) = 260.72553

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.
$43800 + \frac{2}{3} \times 260.72553 = 43973.82$
$43800 - \frac{2}{3} \times 260.72553 = 43626.18$

Interpret your 50% confidence interval in the usual way, using the context:

We are 50% confident that the mean in our populations income is between $43626 and $43973.

Group 4 - Question 1

In a random sample of 765 adults in the United States, 322 say they could not cover a $400 unexpected expense without borrowing money or going into debt. (We have seen this example before.)

Estimate the standard error for this kind of random sample:

.0178 = ((.4209*(1-.4209))/765)

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.
322/765 = .4209
± ⅔ *SE
(.4090,.4328)

Interpret your 50% confidence interval in the usual way, using the context:

We are 50% confident that the interval between .4090 and .4328 includes the population of American adults who cannot cover an unexpected $400 dollar expense without issue.

Question 2

Another example we have been using is about estimating the mean income in a population using the mean income in a random sample of 500 incomes from that population. The sample mean income is $43800, and the sample standard deviation is $5830.

Estimate the standard error for this kind of random sample:

$260.73

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.

X_bar = $43800
+/- (⅔)($260.73) =$173.82
($43626.18, $43973.82)

Interpret your 50% confidence interval in the usual way, using the context:

We are 50% confident that the the mean income of our population lies between $43,626.18 and $43,973.82.

Group 5 - Question 1

In a random sample of 765 adults in the United States, 322 say they could not cover a $400 unexpected expense without borrowing money or going into debt. (We have seen this example before.)

Estimate the standard error for this kind of random sample:

Sample proportion: .4209
q: .5791
n: 765

Standard error: .0178

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.
0.4209 + ⅔ * 0.0178 = 0.43278169934
0.4209 - ⅔ * 0.0178 = 0.40904836601

Interpret your 50% confidence interval in the usual way, using the context:
We are 50% confident that the interval 0.40904 to 0.43278 contains the proportion of all adults in the United States that would say they could not cover a $400 unexpected expense without borrowing money or going into debt.

Question 2

Another example we have been using is about estimating the mean income in a population using the mean income in a random sample of 500 incomes from that population. The sample mean income is $43800, and the sample standard deviation is $5830.

Estimate the standard error for this kind of random sample:

Error: $260.7255

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.

$43800 + \frac{2}{3} * 260.7255 = 43973.817$

$43800 - \frac{2}{3} * 260.7255 = 43626.183$

Interpret your 50% confidence interval in the usual way, using the context:

We are 50% confident that the interval $43,626.183 to $43,973.817 contains the population's mean income.

Group 6 - Question 1

In a random sample of 765 adults in the United States, 322 say they could not cover a $400 unexpected expense without borrowing money or going into debt. (We have seen this example before.)

Estimate the standard error for this kind of random sample:
765 = n
322/765 = 42.09 = p

$\sqrt{(p(1-p)/n)}$
$\sqrt{((.4209(1 - .4209))/ 765)} = 0.0178$

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.

$.4209 + \frac{2}{3} * 0.0178 = .432$
$.4209 - \frac{2}{3} * 0.0178 = .409$

Interpret your 50% confidence interval in the usual way, using the context:

We are 50% confident that the true proportion of americans who say couldn't cover a 400$ unexpected expense is between .409 and .432

Question 2

Another example we have been using is about estimating the mean income in a population using the mean income in a random sample of 500 incomes from that population. The sample mean income is $43800, and the sample standard deviation is $5830.

Estimate the standard error for this kind of random sample:
5830

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.
5830 + $\frac{2}{3}$ *
5830 - $\frac{2}{3}$ *

Interpret your 50% confidence interval in the usual way, using the context:

Group 7 - Question 1

In a random sample of 765 adults in the United States, 322 say they could not cover a $400 unexpected expense without borrowing money or going into debt. (We have seen this example before.)

Estimate the standard error for this kind of random sample:
√p(1-p)/n
√0.46(1-0/46)/765
0.00089

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.
0.46 +-0.00059 = (0.48941, 0.49059)

Interpret your 50% confidence interval in the usual way, using the context:
If we considered many random samples of 765 Americans, and we calculated 50% confidence intervals for each, 50% of these intervals would include the true population proportion of Americans who can't cover an unexpected $400 expense without borrowing money or going into debt

Question 2

Another example we have been using is about estimating the mean income in a population using the mean income in a random sample of 500 incomes from that population. The sample mean income is $43800, and the sample standard deviation is $5830.

Estimate the standard error for this kind of random sample:
$\sigma/\sqrt{n}$
$5830/\sqrt{500}$
260.73

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.
5830 +- 173.82 = (5656.18, 6003.82)

Interpret your 50% confidence interval in the usual way, using the context:
If we considered many random samples of 500 Americans, and we calculated 50% confidence intervals for each, 50% of these intervals would include the true population proportion of the mean income of a population

Group 8 - Question 1

In a random sample of 765 adults in the United States, 322 say they could not cover a $400 unexpected expense without borrowing money or going into debt. (We have seen this example before.)

Estimate the standard error for this kind of random sample:
0.017

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.
.431, .408

Interpret your 50% confidence interval in the usual way, using the context:
We are 50% confident that between .408 and .431 of all americans can not cover a $400 unexpected expense without borrowing money.

Question 2

Another example we have been using is about estimating the mean income in a population using the mean income in a random sample of 500 incomes from that population. The sample mean income is $43800, and the sample standard deviation is $5830.

Estimate the standard error for this kind of random sample:

261.43

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.

5,655.88, 6,004.112

Interpret your 50% confidence interval in the usual way, using the context:

We are 50% confident that the mean income for the population will be between $5655.88 and $6,004.11

Group 9 - Question 1

In a random sample of 765 adults in the United States, 322 say they could not cover a $400 unexpected expense without borrowing money or going into debt. (We have seen this example before.)

Estimate the standard error for this kind of random sample:

0.018

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.
(0.403, 0.439)

Interpret your 50% confidence interval in the usual way, using the context:
We are 50% confident that between 40.3% and 43.9% of adults in the US could not cover an unexpected $400 expense.
That the true % of americans who say that couldn't cover a $400 expense is between ____ and _____.

Question 2

Another example we have been using is about estimating the mean income in a population using the mean income in a random sample of 500 incomes from that population. The sample mean income is $43800, and the sample standard deviation is $5830.

Estimate the standard error for this kind of random sample:
260.73

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.
(43,539, 44,060)

Interpret your 50% confidence interval in the usual way, using the context:
We are 50% confident that the average income in our population is between $43,539 and $44,060.

Group 10 - Question 1

In a random sample of 765 adults in the United States, 322 say they could not cover a $400 unexpected expense without borrowing money or going into debt. (We have seen this example before.)

Estimate the standard error for this kind of random sample:

322/765 = 0.421
sqrt(0.421(1-0.421) / 756) = 0.018

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.

0.421 (+/-) 0.012 = (0.409, 0.433)
Interpret your 50% confidence interval in the usual way, using the context:

We are 50% confident that the true percentage of American adults believe they could not cover a $400 unexpected expense without going into debt is between 40.9% and 43.3%

Question 2

Another example we have been using is about estimating the mean income in a population using the mean income in a random sample of 500 incomes from that population. The sample mean income is $43800, and the sample standard deviation is $5830.

Estimate the standard error for this kind of random sample:

5830 / sqrt(500) = 260.73

Use your standard-error estimate to calculate a 50% confidence interval for the appropriate population parameter.

⅔ (260.73) = 173.82
43800 (+/-) 173.82 = (43626.18, 43973.82)

Interpret your 50% confidence interval in the usual way, using the context:

We are 50% confident that the true mean income is between $43626.18 and $43973.82